

BANARAS HINDU UNIVERSITY
B.A./B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. STA/STB-604: Reliability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Explain the various censoring schemes (Type-I, Type-II, and random censoring) and obtain the corresponding likelihood function for each, with examples. [7]
- (b) Write short notes on:
 - (i) Conditional probability of failure
 - (ii) System configuration (series, parallel, k -out-of- m)
 - (iii) Hazard rate

[7]

Question 2.

- (a) What do you mean by aging of a system? Discuss various failure-time distributions based on the aging effect (IFR, DFR, IFRA) with examples. [7]
- (b) Obtain the maximum likelihood estimate of the survival function and hazard rate in the case of Type-I censoring, where the probability density function is given by:

$$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0, \quad \theta > 0.$$

[7]

Question 3.

- (a) Show that for a coherent system, $h(\mathbf{p})$ is strictly increasing in each component reliability p_i . Also explain that for an n -component series system, the reliability importance of the most reliable component is the least, and for a parallel system the opposite holds. [7]
- (b) Obtain the UMVUE of the survival function for $f(x; \theta) = \theta e^{-\theta x}$, $x \geq 0$, $\theta > 0$, in the case of a complete sample. [7]

Question 4.

- (a) For the exponential distribution $F(x; \theta) = 1 - e^{-x/\theta}$, $x > 0$, $\theta > 0$, show that under Type-II censoring with r ordered observations $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(r)}$, the statistic:

$$T = \sum_{i=1}^r X_{(i)} + (n - r)X_{(r)}$$

is the complete and sufficient statistic for θ . Hence obtain the UMVUE of θ . [7]

- (b) Prove the following results:

(i) $R(t) = \exp\left(-\int_0^t h(x) dx\right)$, where $h(x)$ denotes the hazard rate.

(ii) For a series system of n independent components: $R_{\text{sys}}(t) = \prod_{i=1}^n R_i(t)$.

[7]

Question 5.

Obtain the reliability function for a parallel structure of n independent components where $r_i(t)$ denotes the reliability of the i -th component. Hence compute the reliability of a parallel system when components are independently and identically exponentially distributed with parameter λ . [14]

Question 6.

- (a) What do you mean by the structure function of a system? Explain the different structures (series, parallel, k -out-of- m , bridge) that can be constructed using n components. [7]

- (b) Explain the terms coherent structure and irrelevant component. Show that for a coherent system of n components: $h(\mathbf{p}) = \sum_{i=1}^n p_i(1 - p_i)$. [7]

Question 7.

- (a) State the lack-of-memory (memoryless) property and prove it for the exponential distribution. [7]

- (b) What do you understand by paths and cuts of a system? Hence define minimal path sets and minimal cut sets with examples. [7]

Question 8.

- (a) Discuss the Weibull lifetime model along with its uses and properties. [7]

- (b) Consider the exponential distribution $f(x; \theta) = \frac{1}{\theta}e^{-x/\theta}$, $x > 0$, $\theta > 0$, under a Type-II failure-censoring scheme. Obtain the best critical region for testing:

(i) $H_0 : R(t) = R_0(t)$ vs. $H_1 : R(t) = R_1(t)$, where $R_0(t) > R_1(t)$.

(ii) $H_0 : h(t) = h_0(t)$ vs. $H_1 : h(t) = h_1(t)$, where $h_0(t) > h_1(t)$.

Compute the power function and comment on its nature.

[7]